# 4.6 The Fundamental Theorem of Algebra

Name _____ Period ____ Date _____

Algebra II 4.6 (30-minute plan) · Student sheet

**I can…**

- use the Fundamental Theorem of Algebra to find the number of solutions of a polynomial equation.
- find all the zeros of a polynomial function and write polynomials from their zeros.
- use Descartes's Rule of Signs.

**Vocabulary** · fill in the missing word

**Fundamental Theorem of Algebra** — a _____ equation of degree $n > 0$ has exactly n solutions, counting repeated and imaginary solutions

**Complex Conjugates Theorem** — if $a + bi$ is a zero of a polynomial with real _____, so is $a - bi$

**Descartes's Rule of Signs** — the number of _____ real zeros is the number of sign changes in $f(x)$ or less than that by an even number; use $f(-x)$ for negative zeros

## Part 1 · Notes: fill in the blanks

A polynomial equation of degree n has exactly _____ solutions.

Imaginary zeros come in _____ pairs: $a + bi$ and _____.

Descartes's Rule: count the sign changes in _____ and in _____.

## Part 2 · Examples (work them with the class)

### Example 1

How many solutions does $x^3 + 64x = 0$ have? Find them.

### Example 2

Write a polynomial function f of least degree with leading coefficient 1 that has the zeros 3 and $2i$.

## Part 3 · Partner practice

### Partner practice

Write a polynomial function of least degree with leading coefficient 1 that has the zeros 1 (repeated) and 4i.